AMITY UNIVERSITY
UTTAR PRADESH

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY

Project Progress Report

B. Tech (Computer Science and Engineering)

Group No: 148

Project Title: Deep Learning for AI-Generated Image Detection: A Review of CNN, Vision Transformer, and Explainable Approaches

Area: Artificial Intelligence and Machine Learning (Computer Vision)

Academic Session: 2026-27

Project Guide: Dr. Richa Gupta

Details of Project Team:

Programme:- B.TECH CSE		Year/Semester:- 4th Year /7th Semester	
S. No.	Enrollment No.	Name	Signature
1.	A2305223472	Rohit Yadav	
2.	A2305223568	Vishal	
3.	A2305223449	Hardik Mehlawat	

Paper Summary:

Introduction

Image generators such as generative adversarial networks and, more recently, diffusion models have made synthetic imagery cheap to produce and difficult to identify by eye. The practical consequences are already visible: fabricated photographic evidence, misinformation carried by realistic images, impersonation, and the contamination of image corpora scraped from the web that later become training data for other models. Manual inspection does not scale to the volume involved, so automated detection of AI-generated images has become a working requirement rather than an academic exercise.

The technical difficulty is not building a detector that works. Published detectors reach near-perfect accuracy when the images they are tested on come from the same generator they were trained on.

The difficulty is that this accuracy does not survive a change of generator. Much of what a detector learns is a consequence of one generator's design — its upsampling operator, its particular training run, its decoder — so a new generator that changes any of those does not merely present harder examples, it removes the evidence. The literature records the drop directly: a ResNet-50 reported at 99.9% accuracy on its own training generator scores 54.9% on an unseen one, and detectors reaching 92.77% mean accuracy on a standard benchmark reach 65.77% on realistic images collected in the wild. This cross-generator generalisation gap is the central open problem in the field and is the organising idea of this project.

This report covers the first phase of a two-semester project. The Semester VII deliverable is a review paper, which surveys and compares published detection methods and ends by identifying a specific gap for the second phase to fill. Studies were collected under a protocol fixed before collection began: six databases (IEEE Xplore, ScienceDirect, SpringerLink, the ACM Digital Library, arXiv and Google Scholar), eight query seeds covering GAN, diffusion and dataset-specific terms, and inclusion limited to still-image work published between 2019 and 2026 that reports quantitative results and appears either in a peer-reviewed venue or as a well-cited preprint. Video and audio deepfake detection were excluded. The searches returned 101 records; removing 26 duplicates left 75 for title-and-abstract screening, of which 74 were retained — 48 primary studies read in full and entered into a structured extraction table, 11 secondary studies used selectively, and 15 background references covering generators, backbones and explanation methods. The surveyed work is grouped below by the origin of the evidence a detector uses.

1. Handcrafted, Statistical and Frequency-Domain Forensics

Fourteen of the 48 primary studies detect generated images by measuring a property of the signal rather than by learning one end to end. The starting observation is that the repeated upsampling steps inside a generator leave periodic traces that are visible in the frequency spectrum even when nothing is visible in the picture itself. Methods in this group classify the DCT coefficients, the azimuthal average of the Fourier spectrum, co-occurrence matrices of neighbouring pixels, or a per-model fingerprint estimated as the average residual of that model's output. The reported figures are the highest anywhere in the review: 100.00% on FFHQ from DCT coefficients, above 99% from co-occurrence matrices, 99.2% on uncompressed 1024×1024 images, and 100% on a self-assembled face set from as few as 20 labelled samples.

Those figures describe narrow conditions, and the qualifying results appear alongside them almost immediately. The same spectral-decay feature that gives 99.2% at full resolution falls to 88.8% at 256×256 and becomes unusable once the image passes through 85% JPEG compression, because compression discards exactly the high frequencies the method reads. More damagingly, the traces are a consequence of a design choice rather than of synthesis itself: retraining the generators with bilinear upsampling instead of the usual operator takes detection accuracy built on those traces from 84.8–99.8% down to 0–0.3%. Later work in this group replaces hand-chosen filters with a learned frequency front-end, which is more robust in practice but inherits the same underlying dependence on how a particular generator was built.

2. CNN-Based Detection and Transfer Learning

Nineteen studies — the largest group — treat detection as ordinary binary image classification and solve it with a convolutional network pretrained on ImageNet, either frozen as a feature extractor or fine-tuned end to end. The line begins with a broad claim: a ResNet-50 trained only on ProGAN images, with careful data augmentation, averages 90.8 mAP over eleven unseen GAN architectures. Later work raises that with contrastive training, reaching 95.6% accuracy and 0.997 AUC, and with ensembles that deliberately hold a generator out of training, reaching 0.9995 AUC. Patch-level variants restrict the receptive field so that the classifier is forced onto local texture rather than global composition.

The arrival of diffusion models removed most of that margin, and the comparison table records the collapse twice. GAN-trained detectors tested on GLIDE, Latent Diffusion, Stable Diffusion, ADM and DALL-E 2 fall to roughly 60–76% accuracy on uncompressed images and to about 50–61% once those images are compressed and resized. Measured differently, the same loss is an average 15.2% AUROC drop, and it is asymmetric: a diffusion-trained detector still reaches 94.26% average detection probability on GAN images, while a GAN-trained one manages 26.34% in the reverse direction. The problem then repeats inside the diffusion family itself, where a ResNet-50 at 99.9% on its training generator scores 54.9% on another.

The response visible in the most recent rows of this group is to stop giving the classifier the image. Instead the network classifies something derived from it — the gradient map taken from a pretrained model, for 86.3% mean accuracy over eight test generators; the relationships between neighbouring pixels, for 92.2% over 28 generators; or the residual between an image and its own diffusion reconstruction, for 99.9% average accuracy over eight diffusion models. A related idea removes the trained classifier altogether and scores an image by its autoencoder reconstruction distance. One caution recurs across this group and matters for the second phase of this project: fifteen of the 48 studies train on the same ProGAN distribution, so a large share of the published cross-generator evidence measures one particular transfer rather than transfer in general.

3. Transformer and Attention-Based Approaches

Only two of the 48 primary studies belong here, which is itself a finding worth recording. A vision transformer divides an image into patches and models the relationships between them with self-attention, which suits semantic content but carries no obvious advantage when the evidence is a low-level forensic trace. The one study in the surveyed set that compares the two families under genuinely matched conditions trains a ViT-Base and an EfficientNetV2-M on the same data and tests both across fifteen manipulation methods: the CNN is better in-domain at 81.1% accuracy but falls to 47.0–57.0% on methods it has not seen, while the ViT is weaker in-domain and holds steadier at about 62.0%. The general caveat identified in this review is that backbone comparisons in this literature are usually confounded, because a change of architecture normally arrives together with a change of pretraining data and training objective, so the architecture is rarely the only variable. That observation is one of the direct reasons for the design of the second phase.

4. Multimodal and Foundation-Model-Based Detection

Eight studies build on the finding that a large vision-language model trained for something else entirely already separates real images from generated ones. With CLIP's image encoder left frozen,

a nearest-neighbour rule and a simple linear probe gain 15.07 mAP and 25.90% accuracy on unseen diffusion and autoregressive models relative to the CNN baselines of the time. The strategies built on that result read the same frozen features in different ways: a few-shot linear SVM at about 90.0% average AUC over 18 generators, intermediate encoder blocks rather than the final layer at 91.5% accuracy and 98.8 AP over 20 test sets, lightweight adapters at 98% and 95% accuracy on unseen GANs and diffusion models, prompt tuning at 98.06% mAP over 18 unseen generators, and an orthogonal subspace decomposition at 99.41% mAP and 95.19% mean accuracy. These are currently the strongest cross-generator results in the review, and their common property is that the representation itself is never trained on detection data.

5. Explainability in Detection

Explanation analysis appears in only four of the 48 rows. Grad-CAM and Grad-CAM++ are the standard tools: they weight the feature maps of a convolutional layer by the gradient of the class score and produce a heatmap of where the network looked. Applied to the small CNN released with the CIFAKE dataset, Grad-CAM shows that the classifier keys on small imperfections in image backgrounds rather than on the main subject — a useful result, because it exposes a shortcut, and a limited one, because it does not say which property of those pixels is synthetic. The remaining three rows are a patch-level map of which regions are detectable at all, a prototype-based classifier that is interpretable by construction, and a quantitative comparison of five explanation methods on a single detector. Only one of the four also carries a cross-generator test, so on the evidence of the table, explainability and generalisation are studied separately.

6. Datasets, Benchmarks and Evaluation Metrics

Eleven datasets were compared in a single table recording, for each, the introducing study, the year, the scale, the source of the real images, the generators covered and the limitation its own authors state. The line of development runs from the GAN-era ForenSynths protocol — train on ProGAN, test on eleven unseen generators — through the diffusion-era GenImage and ArtiFact collections, to in-the-wild sets such as Chameleon and TWIGMA. Three problems recur. GenImage's real and generated images differ systematically in JPEG compression and image size, so a classifier can score well by reading those properties instead of any forensic trace; removing both biases raises cross-generator accuracy by more than eleven points for both a ResNet50 and a Swin-T. A single-generator dataset such as CIFAKE cannot expose a cross-generator drop at all. And every dataset fixes its generator list at publication, so all of them go stale.

Eight evaluation metrics were then defined as numbered equations — accuracy, precision, recall, F1 score, false positive rate, AUC, average precision and mean average precision — and the review examined how the surveyed studies actually report against them. Roughly 38 quote an accuracy figure, 14 an average precision and 9 an AUC, and about 23 quote an accuracy figure and nothing else. Averages can hide the worst case: one spectrum classifier averaging 97.2% fails outright on a single generator inside its own test set. This is a direct argument for evaluating candidate architectures under one protocol rather than assembling headline figures from separate papers.

7. Comparative Analysis and the Gap Identified

All 48 primary studies were placed in a single comparison table, ordered by year, with eight columns: study, detection approach, model or architecture, dataset used, generators covered, reported accuracy or AUC, whether a cross-generator test is reported, and the limitation the authors themselves state. The last two columns are what this review adds to the conventional comparison, and every count in the analysis was obtained by tallying a column of the table so that it can be re-checked against it. The finished table was verified against the extraction records by script, and all 48 rows matched with no discrepancy in author strings, accuracy figures or cross-generator flags.

The counts give the shape of the field. Cross-generator evaluation is now common practice but not universal: 36 of the 48 studies report one and 12 do not. Twenty cover GAN-era generators only, eight diffusion-family generators only and nineteen cover both. Fifteen share the ProGAN training distribution. Ten name robustness to compression, resizing or other degradation in their own stated limitation, which makes it the most commonly acknowledged weakness in the table. Explanation analysis appears in four rows.

Reading those counts together produces the gap. Several studies do compare architectures under one protocol, and several do produce an explanation analysis, but no row of the table combines all four of the following: several ImageNet-pretrained CNN backbones trained under a single protocol, a Vision Transformer benchmark trained under the same protocol, an analysis of what the trained models attend to, and a cross-generator test. Each element is well established somewhere in the table; the four together are not reported anywhere in it. That combination is the gap this review identifies, and it defines the study designed for the project's second phase: a transfer-learning comparison of ResNet50, DenseNet121, EfficientNet-B0 and VGG19, each in a feature-extraction and a fine-tuning configuration, benchmarked against a Vision Transformer under matched conditions on CIFAKE, with Grad-CAM applied to correct and incorrect predictions and a cross-generator test on a subset of GenImage.

8. Implementation Work Begun on the Project Guide's Instruction

Following the guide's instruction of 15 August 2026 to begin preparing the implementation code ahead of Semester VIII, the experimental pipeline was built, the core experiment matrix was completed and, since the first progress review on 27 August, its three lower-bound runs have been repeated at a higher epoch ceiling. None of what follows appears in the review paper, which reports no experimental results of its own; these numbers belong to the second phase of the project.

The pipeline is configuration-driven: one YAML file per experiment, and every run logs its seed, split file, optimiser, learning rate, batch size, epoch count, parameter counts, hardware string, library versions and wall-clock training time into a results CSV, so that no number is transcribed by hand into a document. CIFAKE, which holds 120,000 images at 32×32 resolution, was split once with seed 42 into 90,000 training, 10,000 validation and 20,000 test images; that split file is committed and is never regenerated, and the official test set is used only for final evaluation. Training ran on the free tier of Kaggle notebooks, and the pipeline was proved end to end on the laptop CPU before any GPU time was spent on it.

The full matrix of five backbones in two transfer-learning modes — ten runs — was completed by 26 August. The three runs that had stopped at the 30-epoch ceiling with validation loss still

improving were then repeated with a 50-epoch ceiling and every other setting unchanged, so the results file now holds 13 runs. Every run used the same Tesla T4 GPU, the same seed, the same split file and its committed batch size, for 30.12 hours of measured GPU time in total: 19.89 hours for the ten-run matrix and 10.23 hours for the three repeats. Taking the repeat in place of the original where one exists, the test accuracies under fine-tuning are ViT-B/16 98.89%, VGG19 97.91%, DenseNet121 97.60%, EfficientNet-B0 97.56% and ResNet50 96.51%. Under feature extraction they are ViT-B/16 94.75%, DenseNet121 93.49%, ResNet50 92.98%, EfficientNet-B0 92.87% and VGG19 90.55%.

Four observations are being carried into the second-phase write-up. The Vision Transformer wins in both modes, which is a direct answer to the CNN-versus-transformer question the project set out to ask; its lead over the best CNN under fine-tuning is about one percentage point, several times the sampling error on 20,000 test images, though from a single seed. VGG19 inverts between the two modes — last under feature extraction and second under fine-tuning, with by far the largest gain of the five — which is consistent with its frozen ImageNet features encoding object semantics that transfer poorly to a task whose evidence is low-level texture. The fine-tuned results of VGG19, DenseNet121 and EfficientNet-B0, between 97.5% and 97.9%, lie within 0.36 percentage points of one another, so on a single seed they will be reported as statistically indistinguishable rather than ranked.

The fourth observation comes from the repeats, which had two different outcomes. EfficientNet-B0 under feature extraction returned the same best epoch and the same accuracy and stopped early on its own, so its original result was already converged. ResNet50 improved in both modes, from 92.83% to 92.98% under feature extraction and from 95.93% to 96.51% under fine-tuning, but its best epochs fell at 49 and 48 of 50 with validation loss still falling slowly, so both rows remain lower bounds. The ceiling was not raised a third time: ResNet50 stays last under fine-tuning by about one percentage point, so its position is not an artefact of the epoch limit, and the write-up will state that it converges more slowly than the other four backbones under the committed training settings.

The next steps are the Grad-CAM analysis across all five models and the cross-generator evaluation on GenImage. Both are built and have been tested locally on a synthetic copy of the GenImage folder layout, and are set up to run as a single evaluation session on Kaggle over the ten selected checkpoints, including an 80-cell cross-generator sweep. The first launch attempt on 23 September was rejected by Kaggle with an authorisation error before any computation began, so this stage is behind the schedule shown in the first progress report (Weeks 6 to 8); the next step is to replace the access credential and relaunch.

Conclusion

The review paper remains complete in draft: an abstract with six keywords, eight sections totalling roughly 9,850 words, a dataset comparison table covering 11 datasets, the master comparison table covering all 48 primary studies, and three figures. Every reference identifier in the bibliography — 74 in total — was opened and checked against its arXiv, Crossref or proceedings record, with the result logged one line per reference. The assembled draft was submitted for the departmental plagiarism check on 15 August 2026 and the result is awaited.

Three conclusions carry into the second phase. Generalisation across generators is a structural problem rather than an engineering inconvenience, because a change of generator design can remove the evidence a detector relies on rather than merely obscure it. The strongest published cross-generator results share one of three properties: a representation that was never trained on detection data, an input derived from the image rather than the image itself, or training data spanning many generators. And cross-generator evaluation and explanation analysis are almost never carried out together, which is precisely the gap this project addresses. What remains for Semester VII is restyling the report into the official Amity template, the plagiarism-check result and any corrections arising from it, the team's own rewrite pass over the drafted prose, and the guide review in the final week, 5 to 9 October. For the second phase, the evaluation stage described above is the immediate step, followed by the planned ablations and the consolidation of all results into tables and figures. On 25 August 2026 the publication plan was revised, with the guide's approval, to a single merged research paper covering both semesters' work; the two Amity project reports are unchanged.

PERT Chart/Schedule of project completion:-

Task	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Scope, title finalisation, search protocol and section skeleton	✓											
Literature collection and title/abstract screening		✓										
Full-text reading of 48 primary studies; extraction table			✓									
Taxonomy finalisation and Section 1 (Introduction)				✓								
Section 2 (Background: GANs, diffusion models, artefacts)					✓							
Section 3 (Taxonomy of detection approaches, 3.1-3.5)						✓	✓					
Section 4 (Datasets) and Section 5 (Metrics) with tables and figures								✓				
Master comparison table (48 studies) and Section 6									✓			
Section 7 (Gap analysis), Section 8 (Conclusion) and Abstract										✓		
Report assembly, reference verification, plagiarism/AI-text check											✓	
Guide review, corrections, final formatting and submission												✓
Implementation environment, CIFAKE pipeline and split file				✓	✓							
Transfer-learning experiment matrix (5 backbones x 2 modes)					✓	✓						
Lower-bound reruns of three runs at a 50-epoch ceiling						✓	✓					
Grad-CAM analysis and cross-generator evaluation (prepared; run pending, originally Weeks 6-8)											✓	

References: Research papers / books / websites etc.:-

[1] I. J. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, "Generative Adversarial Networks," in Advances in Neural Information Processing Systems (NeurIPS), 2014, arXiv:1406.2661.

[2] T. Karras, S. Laine, M. Aittala, J. Hellsten, J. Lehtinen, and T. Aila, "Analyzing and Improving the Image Quality of StyleGAN," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, arXiv:1912.04958.

[3] J. Ho, A. Jain, and P. Abbeel, "Denoising Diffusion Probabilistic Models," in Advances in Neural Information Processing Systems (NeurIPS), 2020, arXiv:2006.11239.

[4] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-Resolution Image Synthesis with Latent Diffusion Models," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022, arXiv:2112.10752.
[5] Y. Chen and J. Zou, "TWIGMA: A dataset of AI-Generated Images with Metadata From Twitter," in Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2023, arXiv:2306.08310.
[6] X. Wang, H. Guo, S. Hu, M.-C. Chang, and S. Lyu, "GAN-generated Faces Detection: A Survey and New Perspectives," in Proceedings of the 26th European Conference on Artificial Intelligence (ECAI), 2023, arXiv:2202.07145.
[7] L. Verdoliva, "Media Forensics and DeepFakes: An Overview," IEEE Journal of Selected Topics in Signal Processing, vol. 14, no. 5, pp. 910–932, 2020, doi: 10.1109/JSTSP.2020.3002101.
[8] L. Lin, N. Gupta, Y. Zhang, H. Ren, C.-H. Liu, F. Ding, X. Wang, X. Li, L. Verdoliva, and S. Hu, "Detecting Multimedia Generated by Large AI Models: A Survey," 2024, arXiv:2402.00045.
[9] X. Zhang, S. Karaman, and S.-F. Chang, "Detecting and Simulating Artifacts in GAN Fake Images," in Proceedings of the IEEE International Workshop on Information Forensics and Security (WIFS), 2019, arXiv:1907.06515.
[10] R. Durall, M. Keuper, and J. Keuper, "Watch your Up-Convolution: CNN Based Generative Deep Neural Networks are Failing to Reproduce Spectral Distributions," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, arXiv:2003.01826.
[11] S. Yan, O. Li, J. Cai, Y. Hao, X. Jiang, Y. Hu, and W. Xie, "A Sanity Check for AI-generated Image Detection," in Proceedings of the International Conference on Learning Representations (ICLR), 2025, arXiv:2406.19435.
[12] S.-Y. Wang, O. Wang, R. Zhang, A. Owens, and A. A. Efros, "CNN-generated images are surprisingly easy to spot... for now," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, arXiv:1912.11035.
[13] J. Ricker, S. Damm, T. Holz, and A. Fischer, "Towards the Detection of Diffusion Model Deepfakes," in Proceedings of the International Conference on Computer Vision Theory and Applications (VISAPP), 2024, arXiv:2210.14571.
[14] R. Corvi, D. Cozzolino, G. Zingarini, G. Poggi, K. Nagano, and L. Verdoliva, "On the detection of synthetic images generated by diffusion models," in Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2023, arXiv:2211.00680.
[15] M. Zhu, H. Chen, Q. Yan, X. Huang, G. Lin, W. Li, Z. Tu, H. Hu, J. Hu, and Y. Wang, "GenImage: A Million-Scale Benchmark for Detecting AI-Generated Image," in Advances in Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2023, arXiv:2306.08571.
[16] K. Chandrasegaran, N.-T. Tran, and N.-M. Cheung, "A Closer Look at Fourier Spectrum Discrepancies for CNN-generated Images Detection," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2021, arXiv:2103.17195.

[17] J. J. Bird and A. Lotfi, "CIFAKE: Image Classification and Explainable Identification of AI-Generated Synthetic Images," IEEE Access, vol. 12, pp. 15642–15650, 2024, doi: 10.1109/ACCESS.2024.3356122.
[18] M. A. Rahman, B. Paul, N. H. Sarker, Z. I. A. Hakim, and S. A. Fattah, "ArtiFact: A Large-Scale Dataset with Artificial and Factual Images for Generalizable and Robust Synthetic Image Detection," in Proceedings of the IEEE International Conference on Image Processing (ICIP), 2023, arXiv:2302.11970.
[19] D. Cozzolino, D. Gragnaniello, G. Poggi, and L. Verdoliva, "Towards Universal GAN Image Detection," in Proceedings of the IEEE International Conference on Visual Communications and Image Processing (VCIP), 2021, arXiv:2112.12606.
[20] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," in Proceedings of the International Conference on Learning Representations (ICLR), 2021, arXiv:2010.11929.
[21] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization," in Proceedings of the IEEE International Conference on Computer Vision (ICCV), 2017, arXiv:1610.02391.
[22] J. Frank, T. Eisenhofer, L. Schönherr, A. Fischer, D. Kolossa, and T. Holz, "Leveraging Frequency Analysis for Deep Fake Image Recognition," in Proceedings of the International Conference on Machine Learning (ICML), 2020, arXiv:2003.08685.
[23] K. Schwarz, Y. Liao, and A. Geiger, "On the Frequency Bias of Generative Models," in Advances in Neural Information Processing Systems (NeurIPS), 2021, arXiv:2111.02447.
[24] N. Yu, L. Davis, and M. Fritz, "Attributing Fake Images to GANs: Learning and Analyzing GAN Fingerprints," in Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2019, arXiv:1811.08180.
[25] F. Marra, D. Gragnaniello, L. Verdoliva, and G. Poggi, "Do GANs leave artificial fingerprints?," in Proceedings of the IEEE Conference on Multimedia Information Processing and Retrieval (MIPR), 2019, arXiv:1812.11842.
[26] R. Corvi, D. Cozzolino, G. Poggi, K. Nagano, and L. Verdoliva, "Intriguing properties of synthetic images: from generative adversarial networks to diffusion models," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2023, arXiv:2304.06408.
[27] J. Ricker, D. Lukovnikov, and A. Fischer, "AEROBLADE: Training-Free Detection of Latent Diffusion Images Using Autoencoder Reconstruction Error," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, arXiv:2401.17879.
[28] T. Dzanic, K. Shah, and F. Witherden, "Fourier Spectrum Discrepancies in Deep Network Generated Images," in Advances in Neural Information Processing Systems (NeurIPS), vol. 33, pp. 3022–3032, 2020, arXiv:1911.06465.
[29] R. Durall, M. Keuper, F.-J. Pfreundt, and J. Keuper, "Unmasking DeepFakes with simple Features," 2019, arXiv:1911.00686.

[30] Y. Jeong, D. Kim, S. Min, S. Joe, Y. Gwon, and J. Choi, "BiHPF: Bilateral High-Pass Filters for Robust Deepfake Detection," in Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2022, arXiv:2109.00911.
[31] Y. Jeong, D. Kim, Y. Ro, and J. Choi, "FrePGAN: Robust Deepfake Detection Using Frequency-level Perturbations," in Proceedings of the AAAI Conference on Artificial Intelligence, 2022, arXiv:2202.03347.
[32] C. Tan, Y. Zhao, S. Wei, G. Gu, P. Liu, and Y. Wei, "Frequency-Aware Deepfake Detection: Improving Generalizability through Frequency Space Learning," in Proceedings of the AAAI Conference on Artificial Intelligence, 2024, arXiv:2403.07240.
[33] U. Ojha, Y. Li, and Y. J. Lee, "Towards Universal Fake Image Detectors that Generalize Across Generative Models," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023, arXiv:2302.10174.
[34] C. Tan, H. Liu, Y. Zhao, S. Wei, G. Gu, P. Liu, and Y. Wei, "Rethinking the Up-Sampling Operations in CNN-based Generative Network for Generalizable Deepfake Detection," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, arXiv:2312.10461.
[35] L. Nataraj, T. M. Mohammed, S. Chandrasekaran, A. Flenner, J. H. Bappy, A. K. Roy-Chowdhury, and B. S. Manjunath, "Detecting GAN generated Fake Images using Co-occurrence Matrices," in IS&T Electronic Imaging: Media Watermarking, Security and Forensics, 2019, arXiv:1903.06836.
[36] Q. Bammey, "Synthbuster: Towards Detection of Diffusion Model Generated Images," IEEE Open Journal of Signal Processing, vol. 5, pp. 1–9, 2024, doi: 10.1109/OJSP.2023.3337714.
[37] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, arXiv:1512.03385.
[38] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, "Densely Connected Convolutional Networks," in Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2017, arXiv:1608.06993.
[39] M. Tan and Q. V. Le, "EfficientNet: Rethinking Model Scaling for Convolutional Neural Networks," in Proceedings of the International Conference on Machine Learning (ICML), 2019, arXiv:1905.11946.
[40] K. Simonyan and A. Zisserman, "Very Deep Convolutional Networks for Large-Scale Image Recognition," in Proceedings of the International Conference on Learning Representations (ICLR), 2015, arXiv:1409.1556.
[41] L. Chai, D. Bau, S.-N. Lim, and P. Isola, "What makes fake images detectable? Understanding properties that generalize," in Proceedings of the European Conference on Computer Vision (ECCV), 2020, arXiv:2008.10588.
[42] D. Gragnaniello, D. Cozzolino, F. Marra, G. Poggi, and L. Verdoliva, "Are GAN generated images easy to detect? A critical analysis of the state-of-the-art," in Proceedings of the IEEE International Conference on Multimedia and Expo (ICME), 2021, arXiv:2104.02617.
[43] Z. Liu, X. Qi, and P. Torr, "Global Texture Enhancement for Fake Face Detection in the Wild," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, arXiv:2002.00133.

[44] Y. Ju, S. Jia, L. Ke, H. Xue, K. Nagano, and S. Lyu, "Fusing Global and Local Features for Generalized AI-Synthesized Image Detection," in Proceedings of the IEEE International Conference on Image Processing (ICIP), 2022, arXiv:2203.13964.
[45] S. Mandelli, N. Bonettini, P. Bestagini, and S. Tubaro, "Detecting GAN-generated Images by Orthogonal Training of Multiple CNNs," in Proceedings of the IEEE International Conference on Image Processing (ICIP), 2022, arXiv:2203.02246.
[46] C. Tan, Y. Zhao, S. Wei, G. Gu, and Y. Wei, "Learning on Gradients: Generalized Artifacts Representation for GAN-Generated Images Detection," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 12105–12114, 2023, doi: 10.1109/CVPR52729.2023.01165.
[47] Z. Wang, J. Bao, W. Zhou, W. Wang, H. Hu, H. Chen, and H. Li, "DIRE for Diffusion-Generated Image Detection," in Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023, arXiv:2303.09295.
[48] R. Ma, J. Duan, F. Kong, X. Shi, and K. Xu, "Exposing the Fake: Effective Diffusion-Generated Images Detection," in AdvML-Frontiers Workshop at the International Conference on Machine Learning (ICML), 2023, arXiv:2307.06272.
[49] Y. Luo, J. Du, K. Yan, and S. Ding, "LaRE²: Latent Reconstruction Error Based Method for Diffusion-Generated Image Detection," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, arXiv:2403.17465.
[50] G. Cazenavette, A. Sud, T. Leung, and B. Usman, "FakeInversion: Learning to Detect Images from Unseen Text-to-Image Models by Inverting Stable Diffusion," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, arXiv:2406.08603.
[51] B. Chen, J. Zeng, J. Yang, and R. Yang, "DRCT: Diffusion Reconstruction Contrastive Training towards Universal Detection of Diffusion Generated Images," in Proceedings of the 41st International Conference on Machine Learning (ICML), vol. 235, pp. 7621–7639, 2024, https://proceedings.mlr.press/v235/chen24ay.html.
[52] D. Cozzolino, G. Poggi, M. Nießner, and L. Verdoliva, "Zero-Shot Detection of AI-Generated Images," in Proceedings of the European Conference on Computer Vision (ECCV), 2024, arXiv:2409.15875.
[53] J. Park and A. Owens, "Community Forensics: Using Thousands of Generators to Train Fake Image Detectors," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), pp. 8245–8257, 2025, arXiv:2411.04125.
[54] D. A. Coccomini, R. Caldelli, F. Falchi, C. Gennaro, and G. Amato, "Cross-Forgery Analysis of Vision Transformers and CNNs for Deepfake Image Detection," in Proceedings of the 1st International Workshop on Multimedia AI against Disinformation (MAD), pp. 52–58, 2022, doi: 10.1145/3512732.3533582.
[55] H. Liu, Z. Tan, C. Tan, Y. Wei, Y. Zhao, and J. Wang, "Forgery-aware Adaptive Transformer for Generalizable Synthetic Image Detection," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024, arXiv:2312.16649.
[56] A. Aghasanli, D. Kangin, and P. Angelov, "Interpretable-Through-Prototypes Deepfake Detection for Diffusion Models," in Proceedings of the IEEE/CVF International Conference on

Computer Vision Workshops (ICCVW), pp. 467–474, 2023, doi: 10.1109/ICCVW60793.2023.00053.
[57] L. Baraldi, F. Cocchi, M. Cornia, L. Baraldi, A. Nicolosi, and R. Cucchiara, "Contrasting Deepfakes Diffusion via Contrastive Learning and Global-Local Similarities," in Proceedings of the European Conference on Computer Vision (ECCV), 2024, arXiv:2407.20337.
[58] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark, G. Krueger, and I. Sutskever, "Learning Transferable Visual Models From Natural Language Supervision," in Proceedings of the International Conference on Machine Learning (ICML), 2021, arXiv:2103.00020.
[59] D. Cozzolino, G. Poggi, R. Corvi, M. Nießner, and L. Verdoliva, "Raising the Bar of AI-generated Image Detection with CLIP," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2024, arXiv:2312.00195.
[60] C. Koutlis and S. Papadopoulos, "Leveraging Representations from Intermediate Encoder-blocks for Synthetic Image Detection," in Proceedings of the European Conference on Computer Vision (ECCV), 2024, arXiv:2402.19091.
[61] S. A. Khan and D.-T. Dang-Nguyen, "CLIPping the Deception: Adapting Vision-Language Models for Universal Deepfake Detection," in Proceedings of the ACM International Conference on Multimedia Retrieval (ICMR), 2024, arXiv:2402.12927.
[62] C. Tan, R. Tao, H. Liu, G. Gu, B. Wu, Y. Zhao, and Y. Wei, "C2P-CLIP: Injecting Category Common Prompt in CLIP to Enhance Generalization in Deepfake Detection," in Proceedings of the AAAI Conference on Artificial Intelligence, 2025, arXiv:2408.09647.
[63] Z. Yan, J. Wang, P. Jin, K.-Y. Zhang, C. Liu, S. Chen, T. Yao, S. Ding, B. Wu, and L. Yuan, "Orthogonal Subspace Decomposition for Generalizable AI-Generated Image Detection," in Proceedings of the International Conference on Machine Learning (ICML), 2025, arXiv:2411.15633.
[64] Z. Sha, Z. Li, N. Yu, and Y. Zhang, "DE-FAKE: Detection and Attribution of Fake Images Generated by Text-to-Image Generation Models," in Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS), pp. 3418–3432, 2023, doi: 10.1145/3576915.3616588.
[65] Y.-M. Chang, C. Yeh, W.-C. Chiu, and N. Yu, "AntifakePrompt: Prompt-Tuned Vision-Language Models are Fake Image Detectors," 2023, arXiv:2310.17419.
[66] Z. Liu, H. Wang, Y. Kang, and S. Wang, "Mixture of Low-rank Experts for Transferable AI-Generated Image Detection," 2024, arXiv:2404.04883.
[67] H. Wu, J. Zhou, and S. Zhang, "Generalizable Synthetic Image Detection via Language-Guided Contrastive Learning," IEEE Transactions on Artificial Intelligence, vol. 7, no. 6, pp. 3485–3496, 2026, doi: 10.1109/TAI.2025.3641104.
[68] M. Keita, W. Hamidouche, H. B. Eutamene, A. Taleb-Ahmed, D. Camacho, and A. Hadid, "Bi-LORA: A Vision-Language Approach for Synthetic Image Detection," Expert Systems, vol. 42, no. 2, 2025, doi: 10.1111/exsy.13829.
[69] D. A. Coccomini, A. Esuli, F. Falchi, C. Gennaro, and G. Amato, "Detecting images generated by diffusers," PeerJ Computer Science, vol. 10, pp. e2127, 2024, doi: 10.7717/peerj-cs.2127.

[70] A. Chattopadhyay, A. Sarkar, P. Howlader, and V. N. Balasubramanian, "Grad-CAM++: Generalized Gradient-Based Visual Explanations for Deep Convolutional Networks," in Proceedings of the IEEE Winter Conference on Applications of Computer Vision (WACV), pp. 839–847, 2018, doi: 10.1109/WACV.2018.00097.
[71] K. Tsigos, E. Apostolidis, S. Baxevanakis, S. Papadopoulos, and V. Mezaris, "Towards Quantitative Evaluation of Explainable AI Methods for Deepfake Detection," in Proceedings of the 3rd ACM International Workshop on Multimedia AI against Disinformation (MAD), 2024, arXiv:2404.18649.
[72] P. Grommelt, L. Weiss, F.-J. Pfreundt, and J. Keuper, "Fake or JPEG? Revealing Common Biases in Generated Image Detection Datasets," in Proceedings of the European Conference on Computer Vision Workshops (ECCVW), 2024, arXiv:2403.17608.
[73] Y. Hong, J. Feng, H. Chen, J. Lan, H. Zhu, W. Wang, and J. Zhang, "WildFake: A Large-Scale and Hierarchical Dataset for AI-Generated Images Detection," in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, no. 4, pp. 3500–3508, 2025, doi: 10.1609/aaai.v39i4.32363.

Signature(s) of project team:

Rohit Yadav(A2305223472) Vishal(A2305223568) Hardik Mehlawat(A2305223449)

Signature of project guide

Dr. Richa Gupta

Date: